

Project Design Phase

Solution Architecture

Date	30 September 2026
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	3 Marks

Solution Architecture:

The proposed solution uses the ServiceNow platform to provide secure, role-based and record-level access control. The architecture connects users, ServiceNow authentication, user roles, the custom u_institution_details table and script-controlled Access Control Lists (ACLs).

When a user requests access to an institution record, the ServiceNow ACL evaluates the user's assigned role and the required access condition. For READ access, the Branch condition is set to **EEE**, and users with the required **bb1** role can view the permitted **EEE** records. Administrators are given full access.

The architecture also provides separate permissions for different record operations. The **bb2** role controls CREATE access, **bb3** controls WRITE access, and **bb4** controls DELETE access. This allows permissions to be separated according to the user's responsibility.

Architecture Components:

- ✓ **User Layer** – Users and administrators access the ServiceNow application.
- ✓ **Authentication & Role Layer** – ServiceNow identifies the user and checks assigned roles.
- ✓ **ACL Security Layer** – Script-controlled ACLs evaluate the user's role and access operation.
- ✓ **Application/Data Layer** – The u_institution_details table stores institution/student records.
- ✓ **Access Decision Layer** – The system grants or denies access according to the configured ACL rules.

Example – Solution Architecture Diagram:

[User / Administrator → ServiceNow Platform → User Authentication & Roles → Script-Controlled ACL → Role + Branch / Operation Validation → u_institution_details Table → Access Granted / Access Denied]

